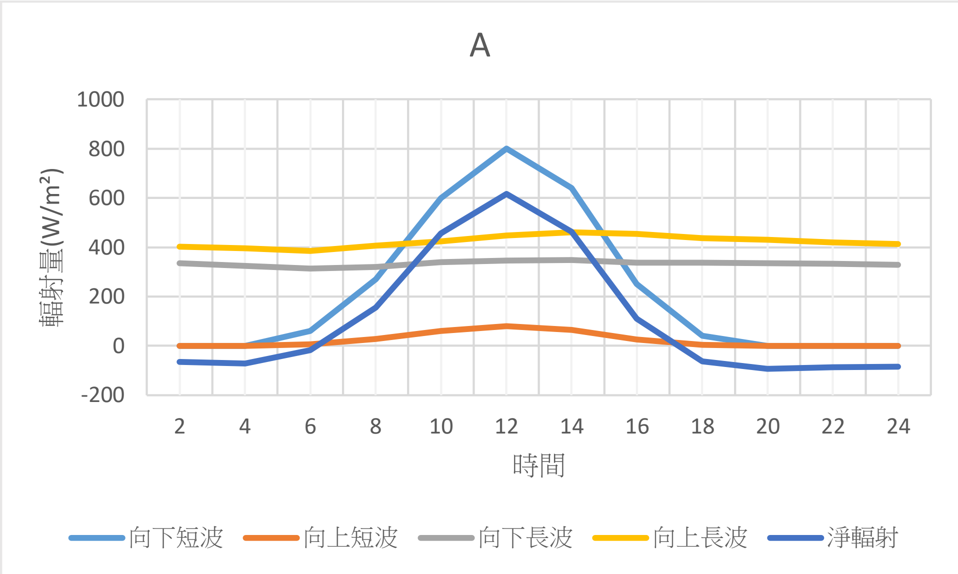
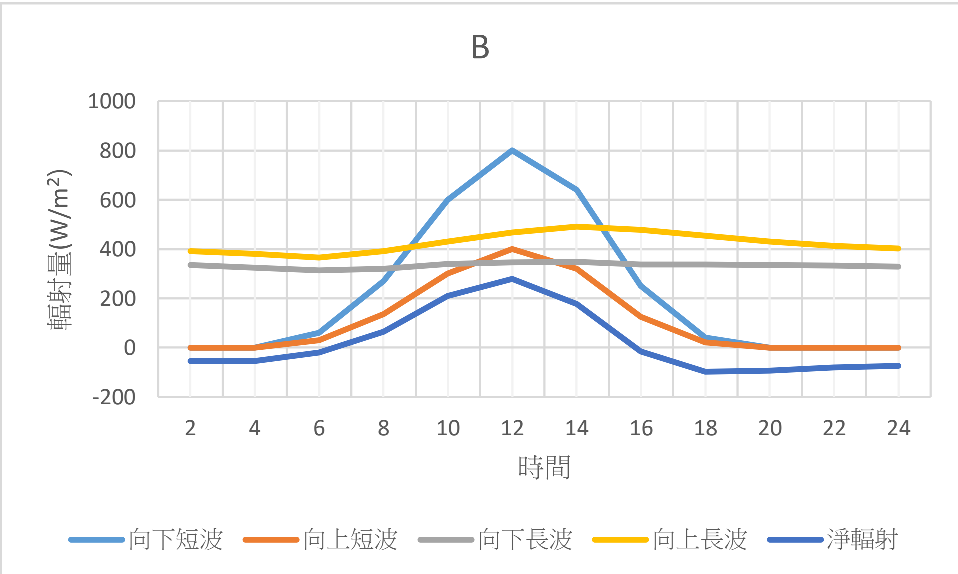
黃若綺

Introduction to Physical Geography Homework　　　　地理一

Question 1　　　　B14208015

1. Please plot the time-series curves of the 5 radiation components (downward shortwave, upward shortwave, downward longwave, upward longwave, and net radiation) as illustrated in the class on two separate figures for these two stations.





2. Please calculate (a) net shortwave radiation, (b) net longwave radiation, and (c) net radiation every two hours in these two stations separately.

時間	A(a)	B(a)	A(b)	B(b)	A(c)	B(c)
2	0.0	0.0	-65.86	-54.89	-65.86	-54.89
4	0.0	0.0	-71.35	-55.15	-71.35	-55.15
6	54.0	30.0	-71.49	-50.46	-17.49	-20.46
8	243.0	135.0	-87.43	-70.89	155.57	64.11
10	540.0	300.0	-84.48	-90.28	455.52	209.72
12	720.0	400.0	-103.05	-121.35	616.95	278.65
14	576.0	320.0	-112.19	-143.63	463.81	176.37
16	225.0	125.0	-116.09	-140.87	108.91	-15.87
18	36.0	20.0	-99.14	-117.09	-63.14	-97.09
20	0.0	0.0	-94.28	-94.28	-94.28	-94.28
22	0.0	0.0	-86.74	-91.05	-86.74	-81.05
24	0.0	0.0	-85.05	-73.86	-85.05	-73.86

（設水泥反照率為 0.5，針葉林反照率為 0.1，忽略日照角度對反照率的影響，σ常數為 $5.67*10^{-8}$）

3. Please discuss the results from your calculation and figures abovementioned. (200 to 300 words).

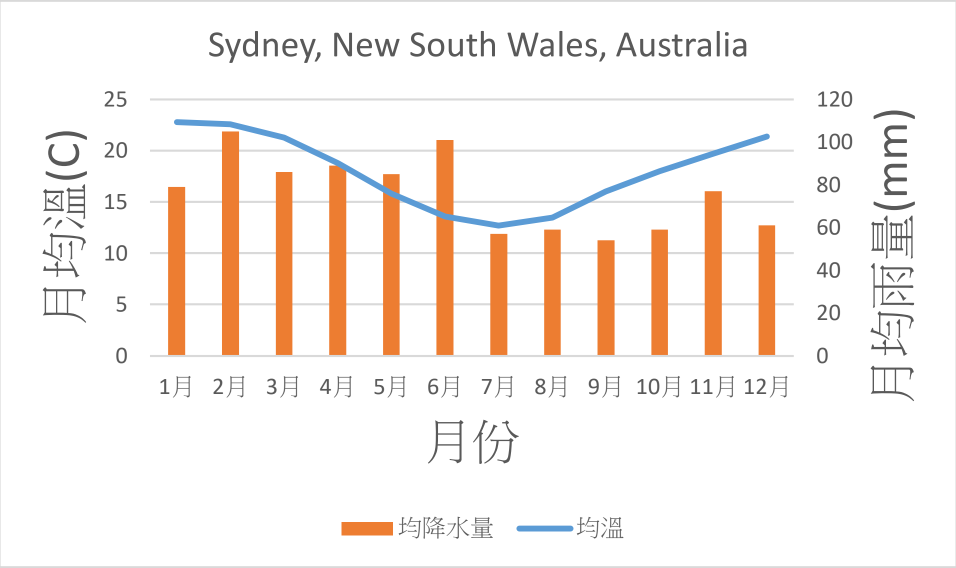
站 A（針葉林區域）因反照率較低（0.1），在白天吸收更多短波輻射，因此淨短波輻射較高，特別是在陽光最強的時段（如 10:00 至 14:00）。相比之下，站 B（水泥覆蓋區域）因反照率較高（0.5），反射了更多短波輻射，導致其淨短波輻射較低。此外，由於水泥地表的溫度更高，站 B 的上行長波輻射也更大，因此夜間淨輻射的散失更顯著。總體來說，地表性質和反照率影響了兩個氣象站的輻射平衡，站 A 的輻射能量累積在白天較高，而站 B 在夜間散失更多能量。

Question 2 :

Please download the climate data of the following 4 cities from the following website https://en.climate-data.org Plot the climatogram for each city and determine its climate classification based on the Köppen Climate Classification introduced in class (do not use any other data sources). In addition, briefly describe how you arrived at your classification.

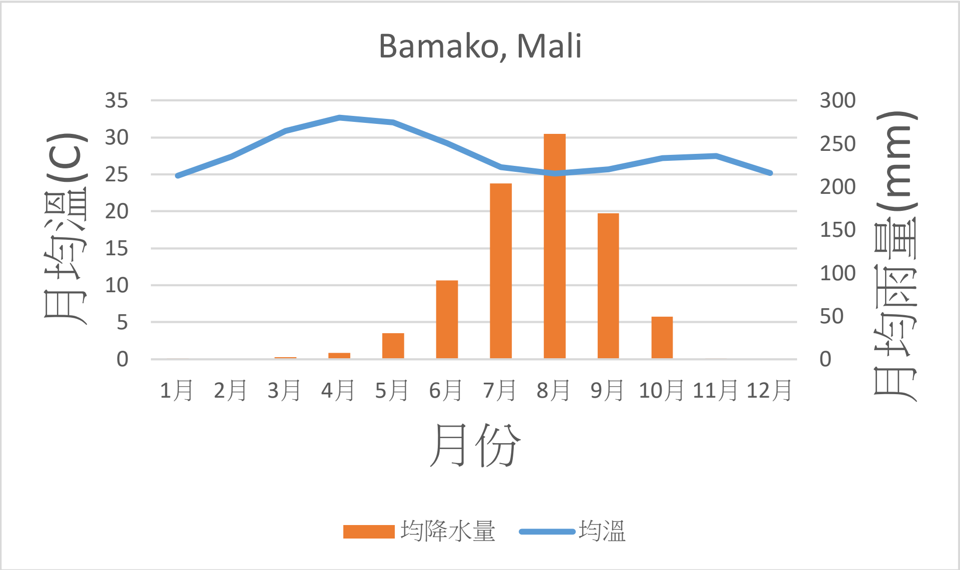
網站有提供 Köppen Climate Classification

1.Sydney, New South Wales, Australia



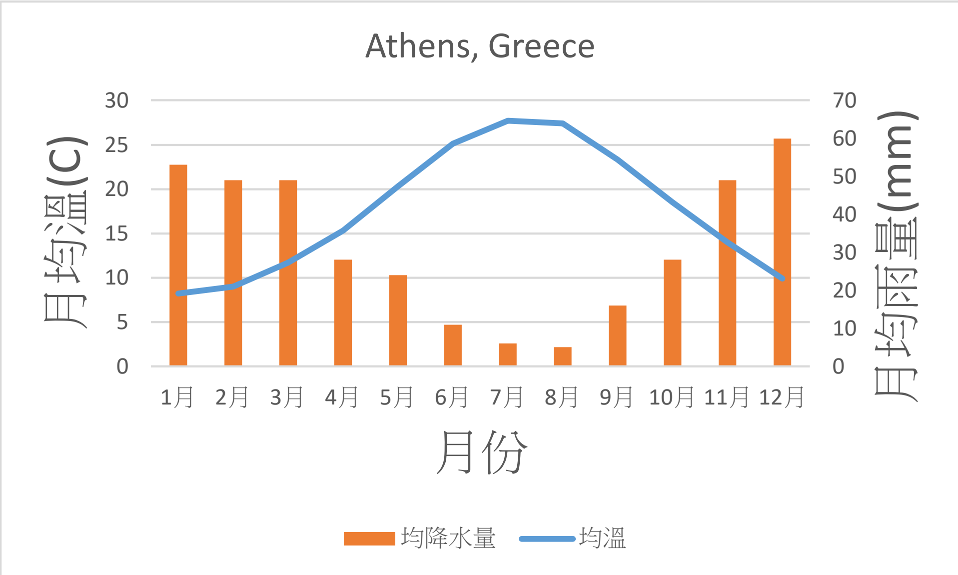
Köppen Climate Classification : Cfa

2.Bamako, Mali



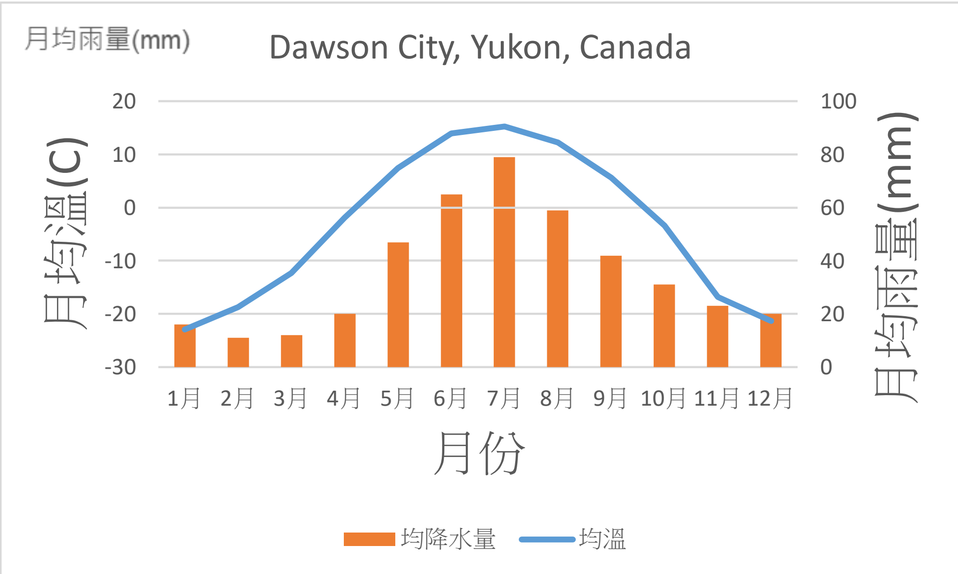
Köppen Climate Classification: aw

3.Athens, Greece



Köppen Climate Classification: Csa

4.Dawson, Yukon, Canada



Köppen Climate Classification: Dfc